BERNOULLI & BINOMIAL DISTRIBUTIONS

Complete Statistics Guide: Definitions, Formulas, Worked Examples, Python Practicals, Advantages, Disadvantages, Comparison and Interview Questions

Learning objectives

- Understand the meaning of Bernoulli and Binomial distributions.
- Learn the probability, mean, variance and standard deviation formulas.
- Solve a worked customer-purchase probability example.
- Build and interpret sample datasets in Python.
- Understand the assumptions, strengths, limitations and interview concepts.

Key idea: Bernoulli models one binary trial; Binomial models the number of successes across a fixed number of independent binary trials.

1. Bernoulli Distribution

Definition

A **Bernoulli distribution** is a discrete probability distribution for one experiment with exactly two possible outcomes. These are commonly coded as Success = 1 and Failure = 0. Success means the event being studied occurs; it does not necessarily mean a good outcome.

Examples: a customer purchases or does not purchase; an email is spam or not spam; a student passes or fails; a visitor clicks or does not click an advertisement.

Formula: probability mass function (PMF)

$$P(X = x) = p^x(1 - p)^{1 - x}, \text{ where } x \in \{0, 1\}$$

Here, X is the outcome, x is the observed value (0 or 1), p is the probability of success, and 1 – p is the probability of failure.

Outcome	Value	Probability
Success	X = 1	P(X = 1) = p
Failure	X = 0	P(X = 0) = 1 – p

Worked example: one customer purchase

Suppose an e-commerce company estimates that a customer has a 70% chance of purchasing a product.

$$p = 0.70 \text{ and } 1 - p = 0.30$$

Outcome	X	Probability
Customer purchases	1	0.70 (70%)
Customer does not purchase	0	0.30 (30%)
Total	—	1.00 (100%)

Mean. variance and standard deviation

$$\text{Mean: } E(X) = p$$

$$\text{Variance: } Var(X) = p(1 - p)$$

$$\text{Standard deviation: } \sigma = \sqrt{[p(1 - p)]}$$

For p = 0.70: mean = 0.70; variance = 0.70 × 0.30 = 0.21; standard deviation = $\sqrt{0.21} \approx 0.4583$.

The mean is the expected value over repeated trials. A single Bernoulli outcome is still only 0 or 1.

Bernoulli practical: customer purchase dataset in Python

The example below creates a small dataset of 10 visitors. The sample contains 7 purchases and 3 non-purchases. Its observed purchase rate is 70%, which happens to equal the assumed theoretical probability in this example.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Assumed probability of purchase
p = 0.70

# Example dataset: 1 = purchase, 0 = no purchase
data = {
    "Customer_ID": range(1, 11),
    "Purchase": [1, 0, 1, 1, 0, 1, 1, 0, 1, 1]
}
```

```
df = pd.DataFrame(data)

print("Customer Dataset:")
print(df)

print("\\nProbability of Purchase:", p)
print("Probability of No Purchase:", 1 - p)

# Observed rate in this sample
purchase_rate = df["Purchase"].mean()
print("Actual Purchase Rate:", purchase_rate)

# Theoretical mean and variance
print("Theoretical Mean:", p)
print("Theoretical Variance:", p * (1 - p))

# Plot theoretical Bernoulli probabilities
outcomes = [0, 1]
probabilities = [1 - p, p]
plt.bar(outcomes, probabilities)
plt.xlabel("Outcome (0 = No Purchase, 1 = Purchase)")
plt.ylabel("Probability")
plt.title("Bernoulli Distribution")
plt.xticks([0, 1])
plt.show()
```

Advantages of Bernoulli distribution

• Simple to understand and calculate.

• Useful for binary events in business, quality checks and classification.

• Can model outcomes such as purchase/no-purchase, spam/not-spam and pass/fail.

• Provides the foundation for the Binomial distribution.

Disadvantages of Bernoulli distribution

• It models only two possible outcomes per trial.

• It describes one trial, not the total successes across many trials.

• The success probability may be unknown and need to be estimated.

• The basic model does not describe dependencies between repeated outcomes.

2. Binomial Distribution

Definition

A **Binomial distribution** calculates the probability of obtaining a specified number of successes in a fixed number of independent trials. Each trial has two possible outcomes, the probability of success remains constant, and the trials are independent.

Examples: number of buyers among 10 customers; number of students who pass in a class; number of defective items in a batch; number of advertisement clicks.

Conditions for a Binomial distribution

• A fixed number of trials, n.

• Each trial has two outcomes: success or failure.

• Trials are independent.

• The probability of success, p, is the same for every trial.

Formula

$$P(X = k) = C(n, k)\, p^{k}(1 - p)^{n - k}$$

P(X = k) is the probability of exactly k successes; n is the number of trials; k is the number of successes; p is the probability of success; and 1 – p is the probability of failure.

$$C(n, k) = n! / [k!(n - k)!]$$

The combination C(n, k) counts the number of ways to arrange k successes among n trials. The factorial symbol (!) means, for example, 4! = 4 × 3 × 2 × 1 = 24.

Worked example: exactly 2 purchases among 5 customers

An online shopping company estimates that each customer has a 30% probability of purchasing. Five independent customers are contacted. Find the probability that exactly two purchase.

Parameter	Value	Meaning
n	5	Total customers / trials
k	2	Required number of purchases
p	0.30	Probability of purchase
1 – p	0.70	Probability of no purchase

$$P(X = 2) = C(5, 2)(0.30)^2(0.70)^3$$

First, C(5, 2) = 5! / [2! × 3!] = 10.

$$P(X = 2) = 10 \times 0.09 \times 0.343 = 0.3087$$

Answer: 0.3087, or **30.87%**. The probability that exactly 2 out of 5 customers purchase is 30.87%.

Mean, variance and standard deviation

$$\text{Mean: } E(X) = np$$

$$\text{Variance: } Var(X) = np(1 - p)$$

$$\text{Standard deviation: } \sigma = \sqrt{[np(1 - p)]}$$

For n = 5 and p = 0.30: mean = 5 × 0.30 = 1.5; variance = 5 × 0.30 × 0.70 = 1.05; standard deviation = $\sqrt{1.05} \approx 1.0247$.

The mean of 1.5 is a long-run average across many groups of five customers. A particular group must have a whole number of purchases.

Binomial practical: generate a dataset in Python

This code simulates 20 groups, with 5 customers in each group. Each customer has a 30% probability of purchase. The simulated values will vary, but the theoretical probability of exactly two purchases remains 30.87%.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import binom

n = 5
p = 0.30

# 20 groups, each with 5 independent customers
np.random.seed(42)
data = np.random.binomial(n=1, p=p, size=(20, n))

df = pd.DataFrame(
    data,
    columns=[f"Customer_{i}" for i in range(1, n + 1)]
)

# Count purchases in each group
df["Total_Purchases"] = df.sum(axis=1)

print("Customer Purchase Dataset:")
print(df)

# Exact probability of 2 purchases
probability = binom.pmf(2, n, p)
```

```
print("\\nProbability of Exactly 2 Purchases:", probability)
print("Percentage:", probability * 100, "%")

# Theoretical mean and variance
print("Theoretical Mean:", n * p)
print("Theoretical Variance:", n * p * (1 - p))

# Probabilities for 0 through 5 purchases
x = np.arange(0, n + 1)
probabilities = binom.pmf(x, n, p)

print("\\nBinomial Probability Distribution:")
for purchases, prob in zip(x, probabilities):
    print(f"{purchases} purchases: {prob:.4f} ({prob * 100:.2f}%)")

plt.bar(x, probabilities)
plt.xlabel("Number of Purchases")
plt.ylabel("Probability")
plt.title("Binomial Distribution (n=5, p=0.30)")
plt.xticks(x)
plt.show()
```

How to read the dataset

Customer_1 ... Customer_5	Each column represents one customer; 1 = purchase and 0 = no purchase.
Total_Purchases	The total number of purchases in that group of five customers.

The dataset rows are random simulation results. Any illustrative row such as 0, 1, 0, 0, 1 has two purchases; the actual generated dataset is determined by the code.

Probability table for n = 5 and p = 0.30

Number of purchases (k)	Probability	Percentage
0	0.16807	16.807%
1	0.36015	36.015%
2	0.30870	30.870%
3	0.13230	13.230%
4	0.02835	2.835%
5	0.00243	0.243%
Total	1.00000	100.000%

The probability of exactly one purchase is 36.015%; the probability of exactly two purchases is 30.870%; and the probability that all five customers purchase is only 0.243%.

Useful SciPy functions: PMF and CDF

```
from scipy.stats import binom

# Probability of exactly 2 successes
print(binom.pmf(2, 5, 0.30))

# Probability of at most 2 successes (0, 1, or 2)
print(binom.cdf(2, 5, 0.30))
```

PMF: `binom.pmf(k, n, p)` calculates the probability of exactly k successes. **CDF:** `binom.cdf(k, n, p)` calculates the probability of k or fewer successes.

Advantages of Binomial distribution

• Models a specified number of successes across multiple trials.

• Its parameters and results are straightforward to interpret.

• Useful in customer purchase, advertisement click and product defect analysis.

• Supports quality control, hypothesis testing and probability modeling.

• Easy to implement using SciPy functions such as `binom.pmf()` and `binom.cdf()`.

Disadvantages of Binomial distribution

• Each trial must have only two outcomes in the model.

• The number of trials must be fixed.

• The probability of success must remain constant across trials.

• Trials must be independent; dependent outcomes may require another model.

• It is not directly suited to continuous values such as spending amounts.

3. Bernoulli vs. Binomial: Comparison

Feature	Bernoulli	Binomial
What it models	Outcome of one trial	Number of successes in multiple trials
Number of trials	One	Fixed number n
Possible values of X	0 or 1	0, 1, 2, ..., n
Probability parameter	p	p
Mean	p	np
Variance	p(1 – p)	np(1 – p)
Probability formula	p^x(1 – p)^(1 – x)	C(n,k)p^k(1 – p)^(n – k)
Example question	Did one customer purchase?	How many of 10 customers purchased?
Python simulation	np.random.binomial(1, p)	np.random.binomial(n, p)

Key relationship: a Bernoulli distribution is a special case of the Binomial distribution when n = 1.

When should you use each?

Use Bernoulli: when modeling one yes/no event, such as whether one customer purchases.

```
import numpy as np
purchase = np.random.binomial(n=1, p=0.30)
print("Purchase outcome:", purchase)  # 0 or 1
```

Use Binomial: when modeling the count of successes across several trials, such as purchases among 10 customers.

```
import numpy as np
purchases = np.random.binomial(n=10, p=0.30)
print("Total purchases:", purchases)  # integer from 0 to 10
```

These are random simulations. Results can vary between runs; a single simulation is not guaranteed to match the expected value.

4. Interview Questions and Answers

Q1. What is a Bernoulli distribution?

A discrete probability distribution for one experiment with two possible outcomes, commonly coded as 1 for success and 0 for failure.

Q2. What is a Binomial distribution?

A distribution that calculates the probability of a specified number of successes in a fixed number of independent trials, with constant success probability.

Q3. What is the main difference between Bernoulli and Binomial distributions?

Bernoulli models one trial; Binomial models the number of successes across multiple trials.

Q4. What is the mean of a Binomial distribution?

E(X) = np, where n is the number of trials and p is the probability of success.

Q5. What is the variance of a Bernoulli distribution?

Var(X) = p(1 – p).

Q6. What are the conditions for a Binomial distribution?

A fixed number of trials, two outcomes per trial, independent trials, and a constant probability of success.

Q7. Can Bernoulli be considered a Binomial distribution?

Yes. Bernoulli is a special case of Binomial with n = 1.

Q8. Which Python library can calculate Binomial probabilities?

SciPy's scipy.stats module provides binom.pmf() for exact probabilities and binom.cdf() for cumulative probabilities.

5. Final Summary

Bernoulli formula:

$$P(X = x) = p^{x}(1 - p)^{1 - x}, x \in \{0, 1\}$$

Use it for one trial with two possible outcomes.

Binomial formula:

$$P(X = k) = C(n, k)\, p^{k}(1 - p)^{n - k}$$

Use it to calculate the probability of exactly k successes in n independent trials.

Practical memory trick

Bernoulli	Did one customer purchase?
Binomial	How many customers out of 10 purchased?

Both distributions are important foundations for probability, statistical inference and machine learning.